INCH-POUND

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SUPERSEDING
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FEDERAL SPECIFICATION

PIPE, STEEL (COAL TAR ENAMEL OR CEMENT-MORTAR LINING, AND COAL TAR ENAMEL COATED AND WRAPPED)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of lined steel water pipe with outer coating and wrapping.

1.2 Classification. Pipe shall be of the following types, as specified (see 6.2):

Type I - Pipe, steel, cement-mortar lined.
Type II - Pipe, steel, coal tar enamel lined for nonpotable water.

1.2.1 Classes and sizes. Pipe shall be normally classified by maximum water working pressure in increments of 50 pounds per square inch (psi) (see 6.2) and shall range from 100 psi through 1000 psi maximum water working pressure. Sizes shall be 4 inches through 144 inches nominal (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4710

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2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Federal Specifications

QQ-S-781	-	Strapping, Steel, and Seals.
PPP-B-587	-	Boxes, Wood, Wirebound Pallet Type.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Federal Standards

FED-STD-123	- Marking for Shipment (Civil Agencies).
FED-STD-601	- Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of the specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials (ASTM)

ASTM A 36	- Standard Specification for Structural Steel.
ASTM A 53	- Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless.
ASTM A 134	- Pipe, Steel, Electric-Fusion (Arc)-Welded (Sizes NPS 16 and over).
ASTM A 135	- Standard Specification for Electric- Resistance-Welded Steel Pipe.
ASTM A 139	- Standard Specification for Electric-Fusion (Arc)-Welded Steel Pipe (NPS 4 and Over).
ASTM A 283	- Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes and Bars.
ASTM A 570	- Standard Specification for Steel, Sheet and Strip, Carbon, Hot-Rolled, Structural Quality.
ASTM A 572	- Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality.
ASTM C 39	- Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens.
ASTM D 395	- Standard Test Methods for Rubber Property - Compression Set.
ASTM D 2000	- Standard Classification System for Rubber Products in Automotive Applications.
ASTM E 8	- Standard Methods of Tension Testing of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Water Works Association (AWWA)

AWWA C 200	- Steel Water Pipe 6 Inches and Larger.
AWWA C 203	- Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape-Hot Applied.
AWWA C 205	- Cement-Mortar Protective Linings and Coating for Steel Water Pipe 4 Inches and Larger - Shop-Applied.
AWWA C 602	- Cement-Mortar Lining of Water Pipelines - 4 Inches and Larger - in Place.

(Application for copies should be addressed to the American Water Works Association, 6666 West Quincy Avenue, Denver, CO 80235.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Description. The pipe shall consist of a seamless or welded steel cylinder with joints formed integrally with the steel cylinder or with steel joint rings welded to the ends. The pipe shall utilize a self-centering bell and spigot joint with a circular preformed elastomeric gasket constructed so that the joint will be watertight under all conditions of design service; a bell and spigot joint for single or double lap welding; ends prepared for butt welding; or a mechanically coupled joint.

3.2 Materials. Materials shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (48 CFR 23.4) to the maximum extent practicable.

3.2.2 Steel.

3.2.3 Pipe. The steel cylinder shall be in accordance with one of the following pipe specifications or shall be fabricated of steel in accordance with one of the following steel plate or sheet specifications, as specified (see 6.2):

Mill-type steel pipe:

ASTM A 53, Grade A or B
ASTM A 135, Grade A or B
ASTM A 139
AWWA C 200

Steel plate pipe:

ASTM A 134
AWWA C 200

Steel plate or sheet:

ASTM A 36
ASTM A 283, Grade C or D
ASTM A 570, Grade B, C, D, or E
ASTM A 572, Grade 42, 45, 50, 55, or 60

3.2.3.1 Specials and bends. Steel sheets and plates used in the fabrication of specials and bends shall be in accordance with the requirements of 3.2.3.1.

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3.2.3.2 Bells and spigots. When welded joint rings are used, the material for bells and spigots shall be either the steel specified in 3.2.3.1 or rolled mill sections of UNS designation G10120.

3.2.4 Mortar. The mortar used for the lining shall be in accordance with AWWA C 205. Cement, aggregate, and water shall be so mixed and of a consistency to produce a dense homogeneous lining that will adhere firmly to the steel cylinder. The mortar shall be tested in accordance with ASTM C 39 and shall develop compressive strength of not less than 2600 psi at 7 days and 4500 psi at 28 days.

3.2.5 Coating, lining, and wrapping material. Materials for coal tar enamel lining, coating, and wrapping shall be in accordance with AWWA C 203, except that the coal tar primer shall be type A or B, as specified (see 6.2).

3.2.6 Elastomeric gaskets.

3.2.6.1 Bell and spigot joints. Elastomeric gaskets for bell and spigot joints shall be made of synthetic rubber in accordance with ASTM D 2000. The volume change of the rubber after immersion in water for 46 hours, ±0.25 hour, at 158 °F, ±3.6 °F, shall not exceed the range of -2 to +5 percent.

3.2.6.1.1 Change in weight. Change in weight due to immersion in water for 2 days at 158 °F shall not exceed 5 percent.

3.2.6.1.2 Compression set. The compression set shall not exceed 20 percent.

3.2.6.1.3 Tensile strength after aging. The elastomeric compound shall not lose more than 20 percent of the tensile strength due to an accelerated aging test in an air oven for 96 hours at 158 °F, or in an oxygen bomb test for 48 hours at 158 °F, 300 psi.

3.2.6.2 Mechanical couplings. Elastomeric gaskets for mechanical couplings shall be made of an elastomeric compound consisting of first-grade natural rubber, synthetic rubber, or a combination thereof, and shall be in accordance with the following physical requirements when subjected to the test specified in 4.6.4.

3.2.6.2.1 Tensile strength. The tensile strength shall be not less than 1000 psi.

3.2.6.2.2 Elongation. Elongation at rupture shall be not less than 175 percent.

3.2.6.2.3 Change in weight. Change in weight due to immersion in water for 2 days at 158 ^{0}F shall not exceed 5 percent.

3.2.6.2.4 Compression set. Compression set shall be determined on a 0.75-inch-diameter by 0.50-inch thick disk, cut from a 3-inch-diameter plaque, subjected to 600 psi for 48 hours at room temperature. The procedure shall conform to the requirements of ASTM D 395, method A, using the external loading

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device. Compression set shall not exceed 4 percent 30 minutes after removal from the test fixture, and not more than 3 percent 3 hours after removal from the test fixture.

3.2.6.2.5 Tensile strength after aging. The tensile strength of the compound shall be not less than 75 percent of the tensile strength before being subjected to an accelerated aging test in an air oven for 96 hours at 158 °F, or an oxygen bomb test for 48 hours at 158 °F, 300 psi.

3.2.6.2.6 Hardness. The hardness of the compound shall be not less than 70 or more than 80, using the Shore A durometer. The increase over the original Shore durometer reading after the accelerated aging test shall be not more than 10 points.

3.3 Fabrication.

3.3.1 Pipe. Pipe cylinders shall be furnished in accordance with one of the specifications specified in 3.2.3.1.

3.3.1.1 Lengths of pipe. Unless otherwise specified (see 6.2), the pipe shall have laying lengths of not less than 20 feet. Longer laying lengths are acceptable and may be of one continuous length of pipe, or produced by welding two or more lengths together in accordance with one of the methods specified for pipe in 3.2.3.1.

3.3.1.2 Wall thickness. Unless otherwise specified (see 6.2), the wall thickness of the pipe shall be not less than that computed on the basis of having a stress of 50 percent of the yield point of the steel at the design pressure.

3.3.1.3 Hydrostatic pressure. The pipe cylinders, including joint ends, shall be hydrostatically tested as specified in 4.6.2.

3.3.1.4 Tolerances. The outside circumference of pipe shall be in accordance with the tolerances allowed under pipe specifications in 3.2.3.1.

3.3.2 Bell and spigot ends. Bell and spigot ends for rubber gasket joints shall be circular in shape and shall contain the gasket in an annular space so that movement of the pipe or hydrostatic pressure shall not displace the gasket. The gasket in joint closure position shall be compressed without the assistance of water pressure. The gasket shall be the sole sealing element to provide watertightness of the joint. The difference in circumferential measurement between the outside circumference of the spigot and the inside circumference of the bell shall be not greater than 0.200 inch for pipe smaller than 72 inches, or greater than 0.100 inch for pipe 72 inches and larger, as measured at the edge of a compressed gasket at nominal joint closure position. The unlined interior surface of the bell and the uncoated exterior surface of the spigot shall be protected by corrosion-resistant paint in accordance with AWWA C 203. Each end of joints formed integrally with the steel cylinder shall be formed by sizing, swaging, rolling, or by expanding over a mandrel. Welded steel joint rings may be formed by one or more pieces of steel welded together, either by resistance

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welding or by manual arcwelding. After welding, the rings shall be smoothed by grinding the weld smooth and flush with the adjacent surface. Welded joint rings or rolled mill sections shall be cold-expanded so they are sized to the manufacturer's design joint diameter. The joint rings may be lap-welded to the pipe. Bell ends shall be pressed or rolled to the required shape without hammering.

3.3.2.1 Bell and spigot end welding. The longitudinal welds on the inside of the bell end and the outside of the spigot end on each section of pipe shall be ground to the plate surface. The sharp edges and burrs in the pipe sections shall be removed from the inside surface of the bell and the outside surface of the spigot.

3.3.3 Ends for mechanically coupled field joints. Ends for mechanically coupled field joints shall be plain, grooved, or banded, as specified (see 6.2). The outside ends of the plain pipe shall be free of surface defects and shall have the welds ground to the surface for a sufficient distance from the ends to make a watertight joint with the coupling. Grooved or banded ends shall be prepared to fit the type of mechanical coupling to be used.

3.3.4 Long radius curves. Long radius curves, horizontal and vertical, shall be made with welded mitered bends or by taking small angular deflections at the bell and spigot joints. If deflections at the bell and spigoted rubber-gasketed joints are used, the manu-facturer shall furnish a joint that will provide for unsymmetrical closure of spigot into the bell so that the rubber-gasketed joint will be watertight at the maximum working pressure and shall be a 0.75-inch maximum pull from a normal 0.25-inch mortar space on one side of the joint in sizes through 24 inches. For sizes larger than 24 inches, a total of not more than 1-inch joint deflection shall be permitted by reducing the normal 0.50-inch inside mortar space to 0.25 inch and pulling the opposite side of the pipe 0.75 inch from normal closure. The angular deflection of gasketed bell and spigot joints shall in no case exceed 5 degrees.

3.4 Cement mortar lining for type I pipe.

3.4.1 Preparation of steel surfaces. Before lining is applied, the cylinder shall be free of loose scale, dirt, grease, or other foreign substances.

3.4.2 Application of lining. The cement mortar lining shall be applied in the shop by a centrifugal method as provided in AWWA C 205. If lining is applied after the pipe is in place in the field, it shall be applied in accordance with AWWA C 602. The process used in the application of the lining shall produce a smooth, dense, durable surface free from sandpockets, voids, oversanded areas, blisters, and cracked areas.

3.4.3 Lining thicknesses. The cement mortar lining thick- nesses shall be in accordance with AWWA C 205 or AWWA C 602.

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3.4.4 Curing. The pipe lining shall be cured by water curing, steam curing, or a combination of both. Water curing and steam curing may be used interchangeably on a time-ratio basis of 4-hours water curing to 1-hour steam curing.

3.4.4.1 Water curing. When water curing is to be used, the lining shall be kept continually moist by spraying or other means for not less than 7 days. The lining shall not be allowed to dry during the curing period. The pipe ends shall be tightly capped with a waterproof cover immediately following completion of the lining of each section to prevent the escape of moisture from the interior of the pipe. When additional moisture is required to maintain a moist condition, water shall be introduced inside the pipe after the mortar has attained sufficient set so the introduction of water shall not damage the mortar. The waterproof covers shall be kept in place throughout and beyond the curing period, the covers to become a component of the completed pipe section. The pipe lining shall be completely cured before application of the enamel coating. When combination water-and-steam curing is employed, it shall be on the time-ratio basis as provided in 3.4.4. The waterproof covers shall be removed for the period of steam curing but shall be replaced whenever steam curing is interrupted or completed.

3.4.4.2 Steam curing. When steam curing is used, the pipe shall be kept in steam maintained at a temperature of 110 to 150 °F for the time-ratio basis provided in 3.4.4. The pipe shall be brought to the curing temperature at a rate of not more than 30°F per hour, provided that in no case shall the temperature exceed 100 °F within 2 hours after the pipe has reached curing temperature. If the lining is to be steam cured, the inside of each section of pipe shall be kept continually moist during the steam curing period. After curing, the pipe shall be protected from rapid drops in temperature that may damage the lining.

3.4.5 Handling. During the entire period of the application of the mortar lining and the curing thereof, the mortar shall be protected from temperatures below 40 °F and the pipe sections shall be carefully supported and handled to avoid injury to the lining, including cracking or separation from the steel surfaces.

3.5 Surface preparation, priming, coal tar enamel lining, coating, and wrapping.

3.5.1 Surface preparation. Surface preparation shall be in accordance with AWWA C 203.

3.5.2 Priming. Priming shall be in accordance with AWWA C 203.

3.5.3 Coal tar enamel coating. Coal tar enamel coating shall be in accordance with AWWA C 203.

3.5.4 Coal tar enamel lining for type II pipe. Application of coal tar enamel lining shall be in accordance with AWWA C 203.

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3.5.5 Coal tar enamel coating and asbestos coal tar saturated-felt overwrap. The coal tar enamel and asbestos coal tar saturated-felt overwrap shall be in accordance with AWWA C 203.

3.5.6 Handling. During the entire period of the application of coal tar lining and coating, the pipe sections shall be supported and handled to avoid injury to the lining and coating. The pipe shall be stored in a manner to prevent damage to the lining and coating.

3.5.7 Coal tar enamel, fibrous-glass mat and bonded asbestos-felt wrap. When specified (see 6.2), a fibrous glass mat, asbestos-felt system shall be in accordance with AWWA C 203.

3.6 Cleaning, lining, and coating of specials and bends. The results of cleaning, lining, and coating of specials and bends shall be equivalent to the results of similar work on straight sections. If the shape precludes fabrication from previously coated and lined pipe sections, the lining and coating shall be applied in accordance with AWWA C 203. Where practicable, the specials shall be fabricated from pipe which has been previously cement-mortar lined by the centrifugal process.

3.7 Electrical detection of flaws. There shall be no defects evidenced in the pipe coating when tested in accordance with 4.6.6.

3.8 Identification marking. Each length of pipe, special, and bend shall be plainly marked by painting to indicate the nominal size, the class for which it is designed, the name of the manufacturer or trademark, contract number, and the date of acceptance by the contracting officer. Type II pipes shall have the words "DO NOT USE FOR DRINKING WATER" painted on the surface in 0.50-inch high letters.

3.9 Workmanship. All pipe shall be uniform in construction and quality. Welds shall be free from cracks and closely spaced-in-line porosity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4).
b. Inspection of preparation for delivery (see 4.7).

4.3 Inspection lot.

4.3.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all fabricated pipe of the same type, class, and size offered at the same time.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.1.2 Sampling for tests.

4.4.1.2.1 Mortar. Not less than four cylinders from each day's run shall be taken for tests as specified in ASTM C 39.

4.4.1.2.2 Tensile specimens. When required, the number of samples for test specimens shall be one for every multiple of 3000 feet of pipe or fraction thereof. This requirement applies only to pipe cylinders fabricated from plates as provided in 3.2.3.1.

4.4.1.2.3 Elastomeric gaskets. Sampling of gaskets shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.1.2.4 Pipe coating. Sampling of pipe for electrical detection of defects shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

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4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.6.1, 4.6.3, 4.6.4, 4.6.5 and 4.6.6, as applicable. AQL shall be 1.0 percent defective.

4.5 Examination procedure.

4.5.1 Examination. The pipe shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.2).
102. Recovered materials not as specified (see 3.2.2).
103. Steel pipe materials not as specified (see 3.2.3.1).
104. Materials for specials and bends not as specified (see 3.2.3.2).
105. Materials for bells and spigots not as specified (see 3.2.3.3).
106. Materials for coating, lining and wrapping material not as specified (see 3.2.5).
107. Lengths not as specified (see 3.3.1.1).
108. Wall thickness not as specified (see 3.3.1.2).
109. Spigot end not properly coated (see 3.3.2).
110. Spigot end dimensions not as specified (see 3.3.2).
111. Bell end dimensions not as specified (see 3.3.2).
112. Bell and spigot end welding not as specified (see 3.3.2.1).
113. Mechanically coupled field joint ends not as specified (see 3.3.3).
114. Inside diameter not as specified (see 3.3.4).
115. Mortar lining surface cracked, rough, blistered, or contains sandpockets, voids, or oversanded areas (see 3.4.2).
116. Cement mortar lining thickness not as specified (see 3.4.3).
117. Curing not as specified (see 3.4.4.1, 3.4.4.2).
118. Separation of lining from steel surfaces (see 3.4.5).
119. Outside coating thickness not as specified (see 3.5.3).
120. Coal-tar lining or coating rough, blistered, or contains holidays (see 3.5.4).
121. Covers missing or not waterproof (see 3.5.5).
122. Workmanship not as specified (see 3.9).

Minor

201. Identification marking not as specified (see 3.8).

4.6 Tests.

4.6.1 Compressive strength for mortar. Test cylinders shall be cast samples of the mortar used in lining the pipe. The curing of test cylinders shall be in conformity with the curing of the pipe. Cylinders shall be molded and tested in accordance with ASTM C 39, except that the cylinders may be centrifugally cast in 6-inch by 12-inch steel molds and tested as hollow cylinders. Nonconformance to 3.2.4 shall constitute failure of this test.

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4.6.2 Hydrostatic pressure. All steel cylinders shall be subjected to a hydrostatic pressure test in accordance with ASTM A 139. Joint ends shall be completed before hydrostatic testing, or, if cylinder has been hydrostatically tested before completion of the joint ends, an approved length of each joint end (except for end areas unaffected by internal working pressure) shall be given an equivalent hydrostatic test. Leakage of cylinders or joint ends shall constitute failure of this test. In lieu of hydrostatic testing, pipe welds at the joint may be checked by magnetic particle or dye penetrant procedures.

4.6.3 Base metal tests for steel cylinders. The base metal test specimens shall be tested in accordance with ASTM E 8 to determine tensile strength. Cylinders from which test specimens are cut may be patch welded. Nonconformance to 3.2.3.1 shall constitute failure of this test.

4.6.4 Elastomeric gaskets. The elastomeric gaskets shall be tested in accordance with the following methods of FED-STD-601. Failure to comply with any of the applicable requirements of section 3 shall be cause for rejection.

	Method
Hardness, durometer	-3021
Compression set	-3311*
Tensile strength	-4111
Elongation, ultimate	-4121
Change in weight liquid immersion	-6251
Oxygen pressure test	-7111
Air heat test	-7221

*Except for mechanical coupling gaskets (see 3.2.6.2.4).

4.6.5 Specials and bends. Specials and bends shall be subjected to the test for pipe in 4.6.1 and 4.6.3, except that specials and bends made from pipe tested in accordance with 4.6.2 shall not be subject to further hydrostatic tests.

4.6.6 Electrical detection flaws. The pipe coating shall be tested in accordance with AWWA C 203. Evidence of any defects will consti- tute failure of this test, and coating shall be repaired and retested.

4.7 Inspection of preparation for delivery.

4.7.1 Quality conformance inspection of preparation for delivery.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

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126. Containers not as specified for level A or C.
127. Rubber gaskets not preserved as specified for level A or C.

128. Gross weight or size exceeds the limitation of the box specification for level A or C.
129. Box not closed and sealed as specified for level A or C.
130. Pipe not bundled as specified for level A or C.
131. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Bells, spigots, specials, bends, and pipe. The bells, spigots, specials, bends, and pipe require no preservation.

5.1.1.2 Rubber gaskets. Rubber gaskets of like description shall be placed in fiberboard boxes in accordance with PPP-B-636, V3c, style optional, in quantities not to exceed the size and weight limitation of the box specification. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.2 Level C.

5.1.2.1 Bells, spigots, specials, bends and pipe. The bells, spigots, specials, bends and pipe require no preservation.

5.1.2.2 Rubber gaskets. Rubber gaskets of like description shall be preserved as specified for level A, except the box shall be domestic type.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Bells, spigots, specials, and bends. Bells, spigots, specials, and bends up to 12-inch diameter shall be packed in a box conforming to PPP-B-587, type I, class 2. The bells, spigots, specials, and bends in excess of 12-inch diameter shall be shipped loose.

5.2.1.2 Rubber gaskets. The rubber gaskets, preserved as specified in 5.1, shall be packed in a box conforming to PPP-B-601, overseas type, grade B, style optional. Strapping shall be in accordance with the appendix to the box specification.

5.2.1.3 Pipe. Pipe with diameters up to 10 inches, and maximum length of 20 feet, of like kind and size shall be bundled by placing one layer, alternating wide ends with narrow ends, on two 4-by-4-by-48-inch timber separators and strapped with steel strapping conforming to QQ-S-781, class 1, type I or IV,

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finish B, size as applicable. Strapping shall be secured to anchor straps nailed to each end of the separator to prevent movement of pipe and facilitate strapping. Straps and separators shall be placed approximately one-sixth of the length of the bundle from each end. Each like bundle containing a single layer of pipe shall be capable of being stacked and bound to each other. Pipe with diameters exceeding 10 inches or lengths exceeding 20 feet shall be shipped loose.

5.2.2 Level C.

5.2.2.1 Bells, spigots, specials, and bends. Bells, spigots, specials, and bends up to a 12-inch diameter shall be packed in a box in accordance with PPP-B-601, domestic type, or PPP-B-621, class domestic, style optional. Those bells, spigots, specials and bends in excess of a 12-inch diameter shall be shipped loose.

5.2.2.2 Rubber gaskets. The rubber gaskets, preserved as specified in 5.1, shall be packed as specified for level A, except the box shall be domestic type, and the strapping shall be finish A.

5.2.2.3 Pipe. Pipe with diameters up to 10 inches and maximum length of 20 feet, of like kind and size, shall be packed as specified for level A, except the strapping shall be finish A.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, all containers shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, all containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pipe covered by this specification is intended for use in water transmission, distribution, and feeder pipe lines for internal working pressures of 100 to 1000 psi. Local conditions may necessitate further design, installation, and operating considerations such as: intermediate working pressures, transient pressure conditions, pipe bedding, depth of backfill, terrain, and other prevailing conditions. Type II pipe shall not be used for potable water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size of pipe (see 1.2 and 1.2.1).
c. Specification to which pipe shall conform (see 3.2.3.1).
d. Type of coal-tar primer to be used (see 3.2.5).
e. Laying lengths of pipe if other than specified (see 3.3.1.1).

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f. Wall thickness of pipe and design stress if other than specified (see 3.3.1.2).
g. Type of end preparation for field joints (see 3.3.3).
h. When fibrous-glass mat is required (see 3.5.7).
i. Size of lot if other than specified (see 4.3.1).
j. Level of preservation and packing required (see 5.1 and 5.2).
k. Finish, if other than specified (see 5.2.1.3).
l. Whether civil agency or military marking is required (see 5.3).

6.4 Subject term (key word) listing.

Pipe, cement-mortar lined
Pipe, coal-tar enamel coated
Pipe, coal-tar enamel lined
Pipe, steel
Pipe, wrapped

6.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodian
Army - ME

Review Activity
DLA - CS

User Activity
Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES:
HHS - FEC
GSA - FSS

PREPARING ACTIVITY:
Army - ME

Project 4710-A863

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